

THETFORD MINES, QC, Canada

WMO: 714940

Lat: 46.0492N

Lon: 71.2661W

Elev: 1411

StdP: 13.96

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-10.1	-21.4	1.8	-13.4	-16.8	2.3	-8.8	22.8	22.1	20.1	18.8	6.9	270	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	80.9	67.3	78.1	65.7	75.6	64.5	70.2	76.7	68.4	74.2	67.0	72.6	10.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	108.9	73.1	66.5	102.8	71.4	64.8	96.7	70.1	35.1	77.2	33.6	74.4	32.3	72.4	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.7	15.0	-21.5	85.5	4.3	1.6	-24.6	86.6	-27.1	87.6	-29.6	88.5	-32.7	89.7		
(5)			-21.8	73.4	4.3	1.4	-24.9	74.4	-27.4	75.2	-29.8	76.0	-32.9	76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	11.8	13.5	23.1	37.2	51.5	59.3	64.6	62.7	55.8	43.5	31.0	18.5		
	DBStd	20.90	13.10	11.31	11.42	9.17	8.19	6.32	5.33	5.21	7.59	8.44	9.87	11.72		
	HDD50	5339	1185	1021	837	397	81	5	0	0	28	238	572	975		
	HDD65	9436	1650	1441	1300	834	425	192	72	107	288	667	1020	1440		
	CDD50	1513	0	0	2	13	127	283	454	393	203	36	2	0		
	CDD65	135	0	0	0	0	6	20	61	35	13	0	0	0		
	CDH74	896	0	0	0	7	77	161	388	193	70	0	0	0		
	CDH80	104	0	0	0	0	8	15	61	14	6	0	0	0		
	WSAvg	7.3	8.1	8.2	8.4	8.2	7.4	6.6	6.1	5.9	6.1	7.3	7.5	7.6		
	PrecAvg	48.3	3.5	2.9	3.1	3.5	3.7	5.0	5.4	4.7	3.9	4.5	3.7	4.2		
(15) Precipitation	PrecMax	57.6	5.2	4.5	4.9	7.1	6.1	8.7	8.5	8.0	7.7	8.9	5.4	8.7		
	PrecMin	34.9	1.5	1.5	1.1	0.5	1.9	1.7	3.0	1.1	1.8	1.6	1.6	2.0		
	PrecStd	5.7	1.1	0.8	1.0	1.7	1.1	1.7	1.4	1.5	1.3	1.9	1.0	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	47.1	57.8	72.0	81.2	82.2	85.2	81.9	80.1	69.8	60.2	46.9		
		MCWB	45.9	45.3	52.5	57.6	63.3	66.6	70.3	69.5	69.3	60.6	53.7	45.4		
	2%	DB	41.3	38.9	47.4	63.5	75.7	78.4	81.4	78.8	75.7	65.1	54.0	41.4		
		MCWB	39.4	36.6	40.8	51.1	60.7	64.5	68.4	66.9	65.8	57.1	49.4	40.2		
	5%	DB	35.2	34.1	42.7	56.9	70.8	75.2	78.2	75.9	72.4	60.8	49.1	37.3		
		MCWB	34.1	31.8	36.9	47.1	57.5	63.4	66.6	65.4	63.8	54.7	45.4	36.3		
	10%	DB	30.5	30.9	38.8	51.9	66.2	71.5	75.4	73.2	69.0	56.6	45.2	33.6		
		MCWB	29.4	29.4	34.5	43.9	55.3	61.1	64.8	64.0	62.2	52.0	42.3	32.3		
	0.4%	WB	46.1	45.9	52.2	59.8	67.9	70.2	73.5	71.6	70.7	62.8	55.5	45.5		
		MCDB	46.7	47.2	58.0	70.1	75.2	77.2	82.4	79.4	76.7	64.9	57.4	46.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.6	37.1	42.9	53.8	63.5	68.2	70.9	69.1	68.1	59.1	50.7	40.1		
		MCDB	41.0	38.4	45.4	60.9	71.2	74.3	77.4	74.8	73.4	63.2	54.3	40.9		
	5%	WB	33.8	32.3	37.9	48.4	60.5	65.6	68.7	67.4	65.8	56.1	46.4	36.3		
		MCDB	35.2	33.5	41.2	54.2	66.9	71.7	74.6	72.7	70.3	60.1	48.3	37.3		
	10%	WB	29.5	29.5	34.9	44.6	57.3	63.2	67.1	65.8	63.4	52.7	42.4	32.4		
		MCDB	30.6	30.9	38.4	50.9	63.5	68.7	73.1	71.2	67.6	55.9	45.2	33.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	16.7	16.8	16.7	19.5	18.0	17.7	17.2	17.9	14.6	13.2	13.4		
		MCDBR	17.3	19.4	21.0	26.0	25.9	23.2	20.8	20.4	20.9	20.5	19.7	16.0		
		MCWBR	17.2	17.8	15.7	16.6	14.0	11.8	10.0	10.5	11.9	14.1	15.6	16.6		
		MCDBR	17.3	18.1	17.7	22.0	20.6	17.7	17.4	16.7	18.3	18.1	17.6	16.2		
	5% WB	MCWBR	17.6	17.5	15.1	15.9	13.7	11.1	9.7	9.8	12.2	14.5	16.0	16.6		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.269	0.303	0.327	0.350	0.388	0.391	0.376	0.347	0.317	0.291	0.276		
	taud		2.370	2.293	2.322	2.430	2.428	2.305	2.309	2.356	2.437	2.524	2.513	2.355		
	Ebn at Noon		274	290	294	296	291	279	276	276	276	271	259	249		
	Edh at Noon		23	30	34	34	35	40	39	36	30	24	20	21		
(44) All-Sky Solar Radiation	RadAvg		444	726	1113	1397	1575	1740	1721	1510	1208	700	415	302		
	RadStd		65	93	88	135	173	117	159	95	104	62	53	31		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page